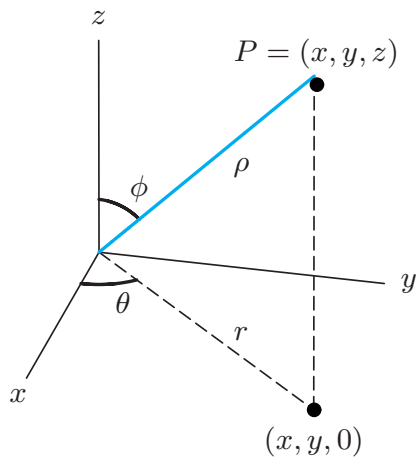
Suggested problems: 1-41 odd

Problem 1: Using the diagram below, write down equations that relate the spherical coordinates (ρ, θ, ϕ) of a point to its cartesian (rectangular) coordinates (x, y, z).



From spherical to cartesian coordinates we have:

$$x = \rho\cos(\theta)\sin(\phi), \quad y = \rho\sin(\theta)\sin(\phi), \quad z = \rho\cos(\phi).$$

We also have $\rho^2 = x^2 + y^2 + z^2$.

Problem 2:

(a) Plot the point with spherical coordinates $(2, \pi/4, \pi/3)$ and find its cartesian coordinates.

See class notes for plot. Cartesian coordinates $= (\sqrt{6}/2, \sqrt{6}/2, 1)$.

(b) Find the spherical coordinates of the point with cartesian coordinates $(0, 2\sqrt{3}, -2)$.

We have $(\rho, \theta, \phi) = (4, \pi/2, 2\pi/3)$.

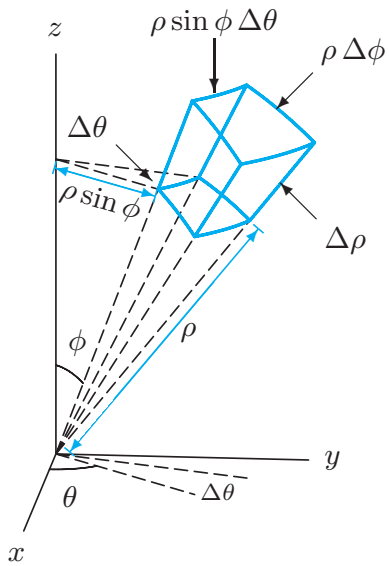
Problem 3:

(a) Describe the equation whose equation in spherical coordinates is $\rho = 1$. Unit sphere centered at the origin.

(b) Give an equation in spherical coordinates for the cone $z = \sqrt{x^2 + y^2}$. $\phi = \pi/4$.

(c) Give an equation in spherical coordinates for the plane $z = 10$. $\rho\cos(\phi) = 10$.

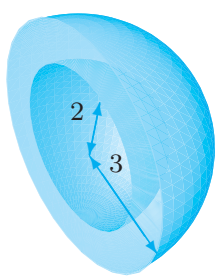
Problem 4: Using the diagram below, express the volume element dV in spherical coordinates.



$dV = \rho^2 \sin(\phi) d\phi d\theta d\rho$

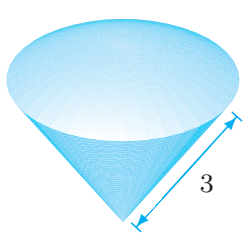
Problem 5: For each object below, set up a triple integral of a function f over the object.

(a)



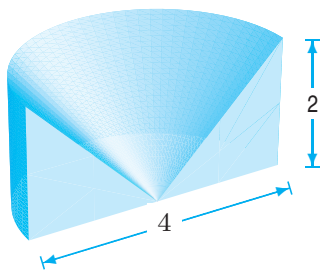
$$\int_0^{\pi} \int_0^{\pi} \int_2^3 f(\rho, \theta, \phi) \rho^2 \sin\phi \; d\rho d\phi d\theta$$

(b) A piece of a sphere, angle at the center is $\pi/3$.



$$\int_0^{\pi/6} \int_0^{2\pi} \int_0^3 f(\rho, \theta, \phi) \rho^2 \sin\phi \; d\rho d\theta d\phi$$

(c)



$$\int_{\pi/4}^{\pi/2}\int_0^{\pi}\int_0^{2/\sin(\phi)} f(\rho,\theta,\phi)\rho^2\sin\phi\ d\rho d\theta d\phi$$

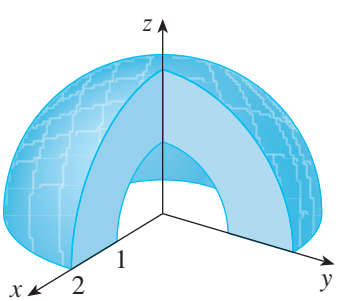
Problem 6: Evaluate

$$\int_W \sin(\phi)\ dV$$

where W is the region $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/4$, and $1 \le \rho \le 2$.

$$\begin{aligned}\int_0^{2\pi}\int_0^{\pi/4}\int_1^2 \sin^2(\phi)\rho^2\ d\rho d\phi d\theta &= \frac{7}{3}\int_0^{2\pi}\int_0^{\pi/4}\sin^2(\phi)\ d\phi d\theta \\ &= \frac{7}{3}\int_0^{2\pi}\int_0^{\pi/4}\left(\frac{1}{2}-\frac{1}{2}\cos(2\phi)\right)\ d\phi d\theta \\ &= \frac{7}{3}\left(\frac{\pi}{8}-\frac{1}{4}\right)\int_0^{2\pi} d\theta \\ &= \frac{14\pi}{3}\left(\frac{\pi}{8}-\frac{1}{4}\right).\end{aligned}$$

Problem 7: Compute the volume of the solid pictured below.



$$\begin{aligned}\int_0^{\pi/2}\int_{\pi/2}^{2\pi}\int_1^2 \sin(\phi)\rho^2\ d\rho d\theta d\phi &= \frac{7}{3}\int_0^{\pi/2}\int_{\pi/2}^{2\pi}\sin(\phi)\ d\theta d\phi \\ &= \frac{7\pi}{2}\int_0^{\pi/2}\sin(\phi)\ d\phi \\ &= \frac{7\pi}{2}\end{aligned}$$

Problem 8: Evaluate

$$\int_W (x^2 + y^2)\ dV,$$

where W lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

$$\begin{aligned}
\int_W (x^2 + y^2)\ dV &= \int_0^{\pi} \int_0^{2\pi} \int_2^3 (\rho^2 \sin^2(\phi))\rho^2 \sin(\phi)\ d\rho d\theta d\phi \\
&= \frac{211}{5} \int_0^{\pi} \int_0^{2\pi} \sin^3(\phi)\ d\theta d\phi \\
&= \frac{422\pi}{5} \int_0^{\pi} \sin^3(\phi)\ d\phi \\
&= \frac{422\pi}{5} \int_0^{\pi} (\sin(\phi) - \sin(\phi)\cos^2(\phi))\ d\phi \\
&= \frac{1688\pi}{15}
\end{aligned}$$